

ENERGY AND ENVIRONMENT

PYQs

Detailed PYQ Solution (2024)

2 SEPTEMBER 2024

Which of the following determines power capacity of a hydropower plant?

A. Length of penstock pipe

B. Flow rate

C. Size of reservoir

D. Head

E. Length from intake to forebay

Choose the most appropriate answer from the options given below:

1. A and C Only

2. B and D Only

3. C and E Only

4. A and E Only

ANSWER: B

Power Capacity of a Hydropower Plant

The power capacity of a hydropower plant is fundamentally determined by two key factors:

1. **Flow Rate (B)**
2. **Head (D)**

1. Flow Rate (B)

The flow rate refers to the volume of water that flows through the hydropower system per unit of time. It is usually measured in cubic meters per second (m^3/s). The higher the flow rate, the more water is available to turn the turbines, which

generates more electricity. Imagine it like the strength of a water stream: the more water you have, the more energy you can extract from it.

2. Head (D)

The head is the height difference between the water source and the water's outflow. It is the vertical distance that the water falls, usually measured in meters. The greater the head, the more potential energy is converted into kinetic energy as the water falls, which in turn spins the turbines more forcefully and generates more electricity. Think of it like dropping water from a greater height, which gives it more force when it hits the turbines.

Remember this formula for future exams

The basic formula for calculating the power generated by a hydropower plant is given by:

$$\text{Power} = \text{Flow Rate} \times \text{Head} \times \text{Gravity} \times \text{Efficiency}$$

Where:

- **Flow Rate** and **Head** directly influence the power output.
- **Gravity** is a constant (approximately 9.81 m/s^2).
- **Efficiency** accounts for losses in the system (like friction and turbine efficiency).

By increasing either the flow rate or the head, you can increase the power output of the hydropower plant. This is why option **2. B and D Only** is the most appropriate choice.

The absorbed dose of radiation dose in SI unit is expressed as

1. Roentgen
2. Rad
3. Gray
4. Becquerel

ANSWER: C

Understanding Radiation Dosage

Radiation refers to the energy emitted from a source and travels through space and may be able to penetrate various materials. When this radiation interacts with matter, it can deposit energy in the material, which may lead to ionization and damage to biological tissues.

Absorbed Dose

The **absorbed dose** is a measure of the amount of radiation energy absorbed by a substance. It specifically quantifies the energy deposited per unit mass of the absorbing material. This measure is critical in radiation protection and medicine because it directly correlates with the potential for biological damage.

Unit: Gray (Gy)

The **Gray (Gy)** is the SI unit for measuring absorbed dose. One Gray is defined as the absorption of one joule of radiation energy per kilogram of matter. Mathematically, it can be expressed as:

$$1\ \text{Gy} = \frac{1\ \text{Joule}}{1\ \text{Kilogram}}$$

To put it into perspective:

- If 1 kilogram of tissue absorbs 1 joule of radiation energy, the absorbed dose is 1 Gray.
- This unit is used because it provides a clear and direct measure of the energy deposited in the tissue, which is crucial for understanding the potential biological effects.

Telegram Channel : Chevish JRF Foundation Whatsapp Number for Inquiry: 8200178822

Which one of the following states in India has the maximum wind power potential at 120 m above ground level as per the recent assessment?

1. Rajasthan
2. Gujarat
3. Karnataka
4. Tamil Nadu

ANSWER: B

Here's the ranking of Indian states by their wind power potential at different heights:

At 100 meters above ground level:

1. **Gujarat**: 123.68 GW
2. **Rajasthan**: 127.75 GW
3. **Karnataka**: 124.15 GW
4. **Maharashtra**: 98.21 GW
5. **Andhra Pradesh**: 74.90 GW

At 120 meters above ground level:

1. **Gujarat**: 142.56 GW
2. **Rajasthan**: 127.75 GW
3. **Karnataka**: 124.15 GW
4. **Maharashtra**: 98.21 GW
5. **Andhra Pradesh**: 74.90 GW

At 150 meters above ground level:

1. **Rajasthan**: 284.25 GW
2. **Gujarat**: 180.79 GW
3. **Maharashtra**: 173.86 GW
4. **Karnataka**: 169.25 GW
5. **Andhra Pradesh**: 123.33 GW

Key Points:

- **Gujarat** consistently ranks high across all heights, making it a leading state in wind power potential.
- **Rajasthan** sees a significant increase in potential at 150 meters, making it the top state at this height.
- **Karnataka** and **Maharashtra** also show strong potential, especially at higher altitudes.
- **Andhra Pradesh** has the lowest potential among the top states but still contributes significantly.

Arrange the following in increasing order of O/C atomic ratio of a coalification path.

A. Vitrinite

B. Huminite

C. Lignin

D. Wood

E. Cellulose

Choose the most appropriate answer from the options given below:

1. A, B, C, D, E

2. B, C, D, E, A

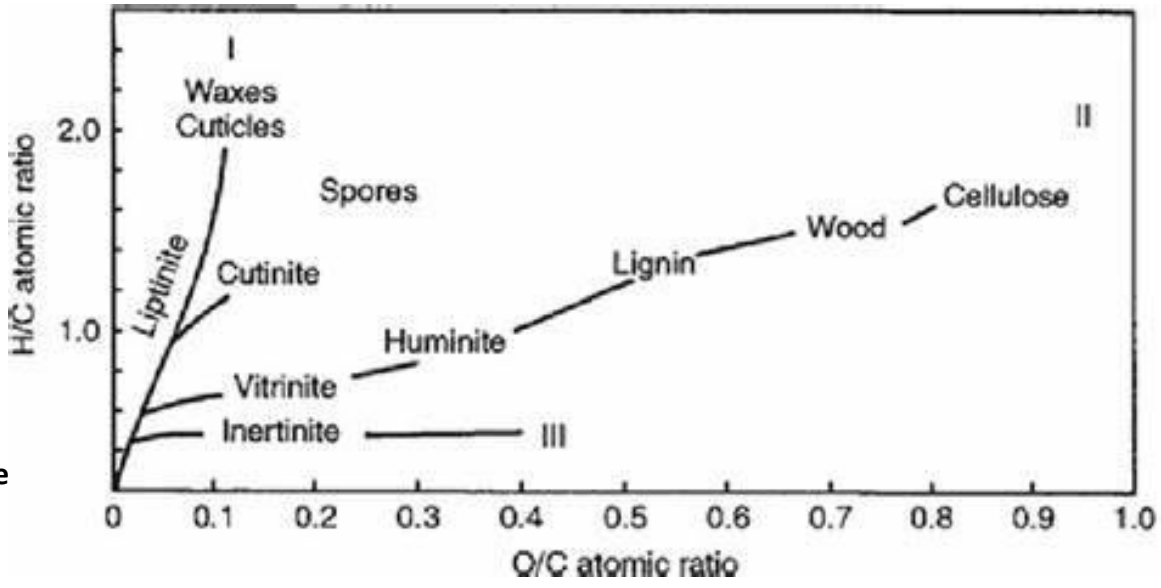
3. D. E, B, C. A

4. E, D, C, B, A

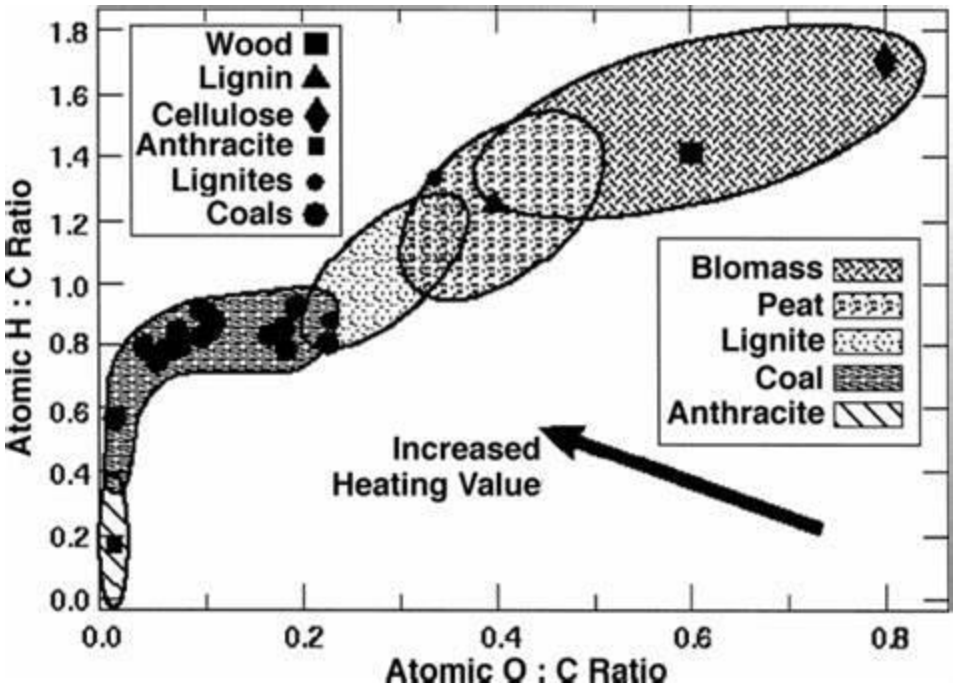
ANSWER: A

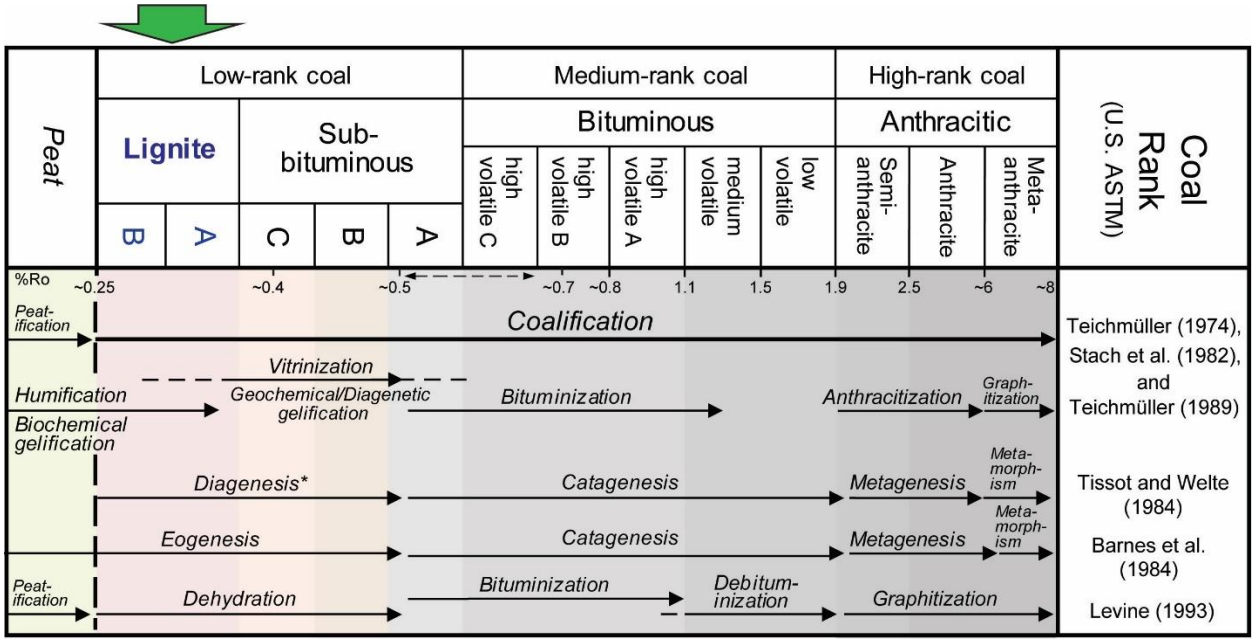
Coalification Path

During the coalification process, organic materials undergo various chemical and physical changes, transforming into different types of coal. The O/C atomic ratio decreases as the material becomes more carbon-rich.



Te





Telegram Channel : Chevish JRF Foundation Whatsapp Number for Inquiry: 8200178822

In biodiesel making, _____ containing sodium hydroxide (lye) solution is mixed with vegetable oil.

1. Glycerol
2. Methanol
3. Ethanol
4. Xylene

ANSWER: B

In the process of making biodiesel, **methanol** containing sodium hydroxide (lye) solution is mixed with vegetable oil. This reaction is called transesterification, where methanol and vegetable oil react in the presence of a catalyst (sodium hydroxide) to form biodiesel and glycerol as a byproduct.

So, the correct answer is:

2. Methanol

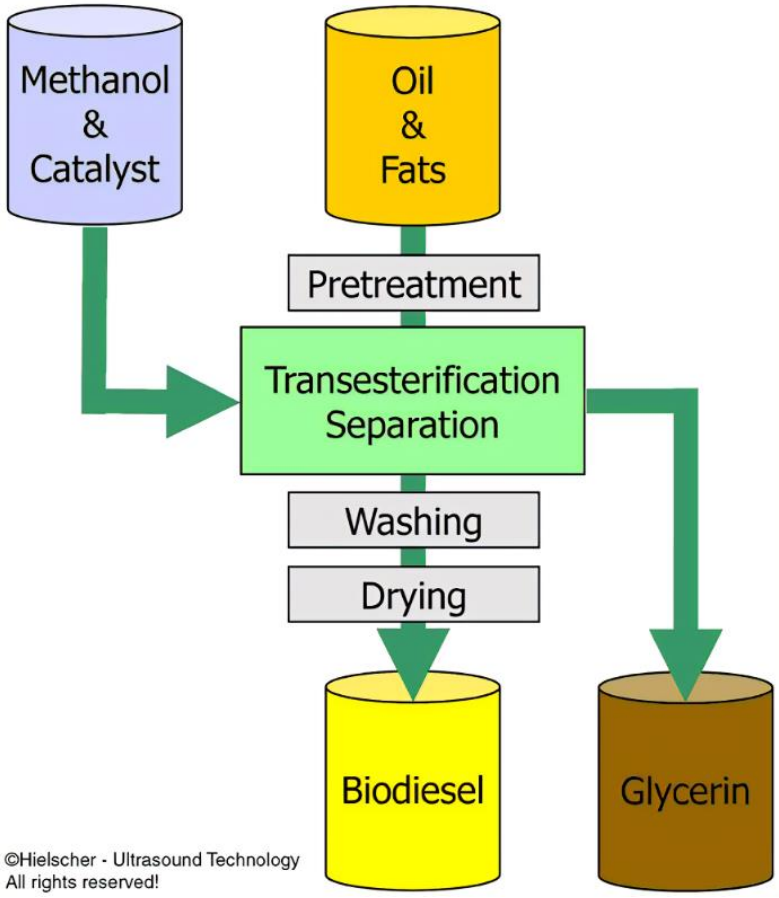
Let's go through the detailed process of biodiesel production step-by-step:

1. Feedstock Preparation

The first step involves selecting and preparing the raw materials, which can be vegetable oils (like soybean, palm, or rapeseed oil), animal fats, or recycled cooking oil. The feedstock is cleaned to remove impurities such as dirt, water, and food particles.

2. Pre-treatment

The feedstock is pre-treated to remove any water and free fatty acids that could interfere with the transesterification process. This is typically done through a process called **degumming** and **neutralization**.



3. Transesterification

This is the core chemical reaction where the triglycerides in the oil or fat react with an alcohol (usually methanol or ethanol) in the presence of a catalyst (usually sodium hydroxide or potassium hydroxide) to form **biodiesel** (fatty acid methyl esters or fatty acid ethyl esters) and **glycerol** as a byproduct.

Transesterification Reaction:

$$\text{Triglyceride} + 3\text{Methanol} \rightarrow 3\text{Biodiesel} + \text{Glycerol}$$

4. Separation

After the reaction, the mixture is allowed to settle. The biodiesel, being lighter, floats on top and is separated from the glycerol, which sinks to the bottom.

5. Washing

The separated biodiesel is washed with water to remove any residual catalyst, soap, and glycerol. This step is crucial to ensure the purity of the biodiesel.

6. Drying

The washed biodiesel is then dried to remove any water content. This can be done using a drying agent or by heating the biodiesel.

7. Purification

Any remaining impurities, such as excess methanol, are removed through a process called **purification**. This ensures that the biodiesel meets the required quality standards.

8. Quality Testing

The final biodiesel product is tested to ensure it meets the required specifications for use as a fuel. This includes testing for viscosity, flash point, and other properties.

9. Storage and Distribution

The finished biodiesel is stored in appropriate containers and can be distributed for use as a fuel in diesel engines.

Advantages of Biodiesel:

- **Renewable**: Biodiesel is made from renewable resources like vegetable oils and animal fats.
- **Biodegradable**: It degrades more quickly than petroleum diesel, reducing environmental impact.
- **Lower Emissions**: Biodiesel produces lower levels of carbon monoxide, particulate matter, and hydrocarbons compared to conventional diesel.

- **Carbon Neutral**: The carbon dioxide released during combustion is offset by the carbon dioxide absorbed during the growth of the feedstock plants.

Disadvantages of Biodiesel:

- **Higher NOx Emissions**: Biodiesel can produce slightly higher nitrogen oxide emissions compared to petroleum diesel.
- **Cold Weather Performance**: Biodiesel has a higher gel point, which can cause issues in colder climates.
- **Compatibility**: Some older engines and fuel systems may not be fully compatible with biodiesel without modifications.

Which one of the following is a thorium mineral used as atomic fuel?

1. Pitchblende
2. Monazite
3. Apatite
4. Malachite

ANSWER: B

Radioactive Element	Mineral	Description
Uranium	**Uraninite (Pitchblende)**	Primary ore of uranium, containing 50-80% uranium oxide (UO_2)
	Carnotite	Potassium uranium vanadate mineral, containing 5-20% uranium
	Autunite	Hydrated calcium uranium phosphate mineral, containing 2-10% uranium
Thorium	**Monazite**	Rare earth phosphate mineral, containing 2-20% thorium oxide (ThO_2)
	Thorianite	Thorium oxide mineral, containing 70-90% thorium oxide
Lithium	**Lepidolite**	Source of lithium for nuclear fusion reactions
	Spodumene	Source of lithium for nuclear fusion reactions
Beryllium	**Beryllium oxide**	Used as a neutron moderator in nuclear reactors
Zirconium	**Zircon**	Used in nuclear reactors due to low neutron absorption cross-section
Potassium	**Orthoclase**	Contains naturally occurring radioactive isotope Potassium-40 (K-40)
	Microcline	Contains naturally occurring radioactive isotope Potassium-40 (K-40)

Telegram Channel : Chevish JRF Foundation Whatsapp Number for Inquiry: 8200178822

Arrange the following regions of India in decreasing order of their identified Hydro power capacity (in MW).

A. Eastern Region

B. North Eastern Region

C. Northern Region

D. Southern Region

E. Western Region

Choose the most appropriate answer from the options given below:

1. C. D. A. B. E

2. D, C, E, B, A

3. B, C, D, A. E

4. C. B. A. D. E

ANSWER: C

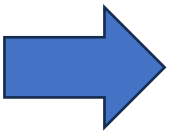
NORTH EASTERN	
Meghalaya	2026
Tripura	0
Manipur	615
Assam	643
Nagaland	325
Arunachal Pradesh	50394
Mizoram	1927
Sub Total (NER)	55930



Region/ State	Identified Potential (MW) (2017-23)
NORTHERN	
Jammu & Kashmir	12265
Ladakh	707
Himachal Pradesh	18305
Punjab	1301
Haryana	0
Rajasthan	411
Uttarakhand	13481
Uttar Pradesh	502
Sub Total (NR)	46971



SOUTHERN	
Andhra Pradesh	2596
Telangana	1302
Karnataka	4414
Kerala	2473
Tamil Nadu	1785
Sub Total (SR)	12570



WESTERN	
Madhya Pradesh	2819
Chhattisgarh	1311
Gujarat	550
Maharashtra	3144
Goa	0
Sub total (WR)	7824



EASTERN	
Jharkhand	300
Bihar	130
Odisha	2825
West Bengal	809
Sikkim	6051
Sub Total (ER)	10115

ion Whatsapp Nu

Which one of the following coalfields belong to lignite type coal deposit?

1. Raniganj, West Bengal
2. Singarauli, Madhya Pradesh
3. Makum, Assam
4. Talcher, Odisha

ANSWER: C

The coalfield that belongs to the lignite type coal deposit is **Makum, Assam**.

Lignite, also known as brown coal, is a low-grade coal with a high moisture content and low heating value, primarily used for power generation

- **Lignite Coal** indicates Tertiary Coalmines **& Coking as well as non-coking and bituminous as well as sub-bituminous coal** are obtained from Gondwana coal fields.

Here's a detailed table listing the major coal mines in India along with the grade of coal they produce:

Coal Mine	State	Coal Grade	Description
Jharia	Jharkhand	Coking Coal	One of the oldest and richest coalfields in India, producing high-quality coking coal for metallurgical purposes.
Giridih	Jharkhand	Coking Coal	Known for producing the finest coking coal in India.
Raniganj	West Bengal	Bituminous Coal	The first coalfield in India, producing bituminous coal used for

Coal Mine	State	Coal Grade	Description
			power generation and industrial purposes.
Korba	Chhattisgarh	Bituminous Coal	Major coalfield producing bituminous coal for power generation.
Talcher	Odisha	Bituminous Coal	Produces coal suitable for steam and gas production, used in thermal power plants.
Singrauli	Madhya Pradesh	Sub-bituminous Coal	Produces sub-bituminous coal used for power generation.
Makum	Assam	Lignite	Produces lignite coal, primarily used for power generation.
Neyveli	Tamil Nadu	Lignite	Produces lignite coal, primarily used for power generation.
Ledo	Assam	Lignite	Produces lignite coal, primarily used for power generation.
Mand-Raigarh	Chhattisgarh	Bituminous Coal	Produces bituminous coal used for power generation and industrial purposes.
Hasdeo Arand	Chhattisgarh	Bituminous Coal	Produces bituminous coal used for power generation and industrial purposes.
Bokaro	Jharkhand	Coking Coal	Produces high-quality coking coal for metallurgical purposes.
South Karanpura	Jharkhand	Bituminous Coal	Produces bituminous coal used for power generation and industrial purposes.

Coal Mine	State	Coal Grade	Description
North Karanpura	Jharkhand	Bituminous Coal	Produces bituminous coal used for power generation and industrial purposes.
Ramgarh	Jharkhand	Bituminous Coal	Produces bituminous coal used for power generation and industrial purposes.
Daltonganj	Jharkhand	Sub-bituminous Coal	Produces sub-bituminous coal used for power generation.
Ib Valley	Odisha	Bituminous Coal	Produces bituminous coal used for power generation and industrial purposes.
Jharsuguda	Odisha	Bituminous Coal	Produces bituminous coal used for power generation and industrial purposes.
Rampia	Odisha	Bituminous Coal	Produces bituminous coal used for power generation and industrial purposes.
Chumukedima	Nagaland	Lignite	Produces lignite coal, primarily used for power generation.
Namchik-Namphuk	Arunachal Pradesh	Lignite	Produces lignite coal, primarily used for power generation.

Arrange the following process from initial to last in formation of a petroleum deposit.

A. Accumulation

B. Generation

C. Seepage

D. Primary migration

E. Secondary migration

Choose the most appropriate answer from the options given below:

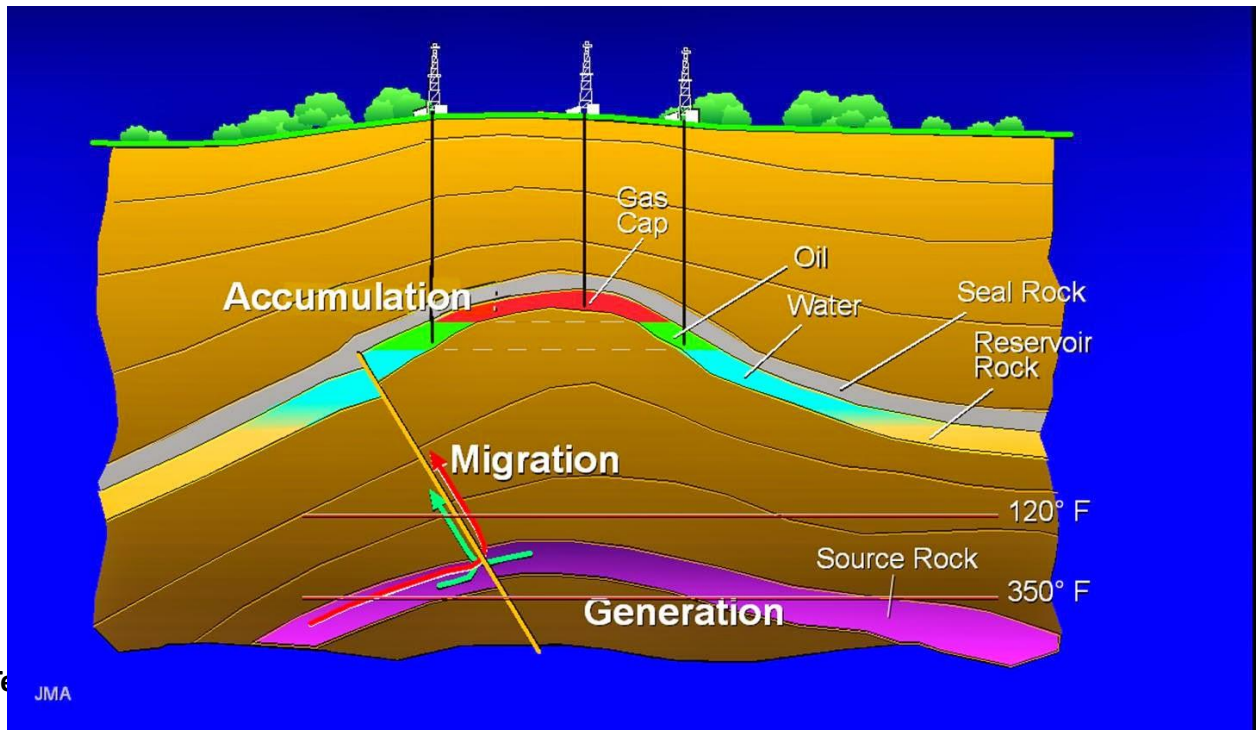
1. A. B. C. D. E

2. B. C. A. D. E

3. B. D. E. A. C

4. A. D. B, E, C

ANSWER: C



5 SEPTEMBER 2024

The half-life of a radioactive isotope of X is 50 years. In a nuclear power plant accident, 1 kg of the X isotope is dispersed into the surroundings of the plant. How much of the X isotope will remain in the surroundings after 150 years?

(1) 500 gm

(2) 250 gm

(3) 125 gm

(4) 12.5 gm

ANSWER: C

Let's calculate how much of the radioactive isotope X will remain after 150 years given that its half-life is 50 years.

Half-Life Calculation

The formula to determine the remaining amount of a radioactive substance after a given time period is:

$$N = N_0\left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$$

Where:

- N is the remaining quantity of the substance.
- N_0 is the initial quantity of the substance.
- t is the time that has elapsed.
- $t_{1/2}$ is the half-life of the substance.

In this case:

- N_0 = 1 kg (1000 grams)
- t = 150 years
- $t_{1/2}$ = 50 years

So, the equation becomes:

$$N = 1000\left(\frac{1}{2}\right)^{\frac{150}{50}}$$

$$N = 1000\left(\frac{1}{2}\right)^{3}$$

$$N = 1000 \times \frac{1}{8}$$

$$N = 125 \text{ grams}$$

r Inquiry: 8200178822

Match List - I with List - II.

LIST-I Petroleum Fraction		LIST-II Numbers of Carbon Atoms	
A.	Gasoline	**I.**	>C25
B.	Diesel	**II.**	C5-C10
C.	Asphalt	**III.**	C15-C25
D.	Kerosene	**IV.**	C10-C18

Choose the correct answer from the options given below :

(1) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

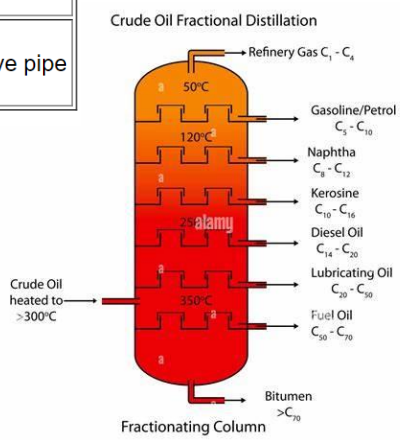
(2) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(4) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

ANSWER: C

Fraction	Number of Carbon Atoms per Molecule	Boiling Point Range ^{o}C	Uses
Natural gas	1- 4	Below 20	Fuel for heating and lighting, for the manufacture of certain compounds, such as ethyne, hydrogen, carbondisulphide, and tetrachloromethane.
Petroleum ethers and ligroin	5 - 7	20 - 120	As organic solvent
Petrol (gasoline)	5 - 12	40 - 205	As motor and aeroplane fuel, for heating and lighting, and as solvent for grease and paints.
Kerosene (paraffin oils)	12 - 18	175 - 325	For heating and lighting; fuel for jet engine, and for driving tractors. Also as solvent for grease and paints.
Gas oil and diesel oil	12 - 25	275 - 400	As fuel for diesel engine, for heating, and as raw materials in the cracking process.
Lubricating oils, waxes, grease, and vaseline	Above 20	Above 400	Lubrication, medicine, manufacture of candle, polish, ointments, creams, and water proofing materials.
Bitumen, asphalt, and other residue	Above 40	Above 400	For surfacing roads and air fields; other residues may be used as fuel, as protective pipe coating, and as paint.



Telegram Channel : Chevish JRF Foundation Whatsapp Number for Inquiry: 82

Cracking of Petroleum

Cracking of petroleum is the breakdown of hydrocarbon fractions (alkanes) with large number of carbon atoms into smaller molecules, or the conversion of straight chain hydrocarbons to branched-chain hydrocarbons.

There are two methods of cracking - thermal (carried out by heating to high temperatures of about 700°C and pressures of about 30 atm) and catalytic (the use of aluminum oxide and silicon(IV) oxides as catalyst).

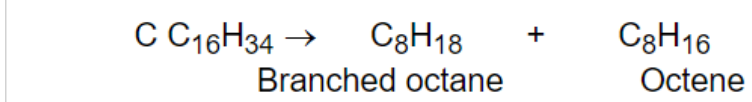
Reason for Cracking

1. **To obtain large quantity of petrol -** i.e., to produce more of small molecular hydrocarbons, which are in the C5 - C12 range. Petrol is high in demand, but is produced from the fractional distillation of crude oil in small quantity.

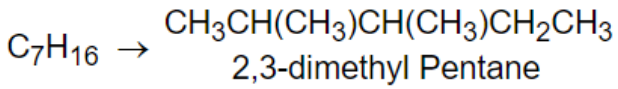
2. **To produce petrol of higher quality** - i.e., to convert straight- chain petrol to branched-chain petrol. The branched-chain petrol burns more smoothly; hence they are more efficient in an internal combustion engine than the straight-chain petrol which is produced more from the fractional distillation of crude oil.

Examples of reactions that occur during cracking process:

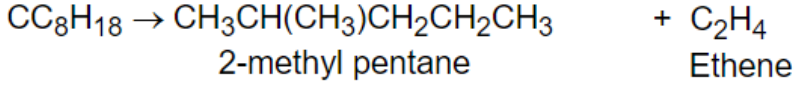
1. Conversion of large molecular alkane to high grade petrol containing branched-chains.



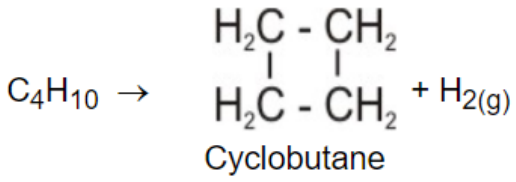
2. Conversion of a straight chain alkane to a branched-chain.



3. Conversion of a straight chain alkane to a branched chain alkane and an alkene.



4. Conversion of a straight chain alkane to a cycloalkane and hydrogen gas



***Note*:** cracking process yields also several gases as by-products, e.g., hydrogen, ethene, benzene, ethane, butene, propene and propane, which can be used in reforming alkanes in the petrol range, and as raw materials in the petrochemical industry for the manufacture of several useful chemicals and products.

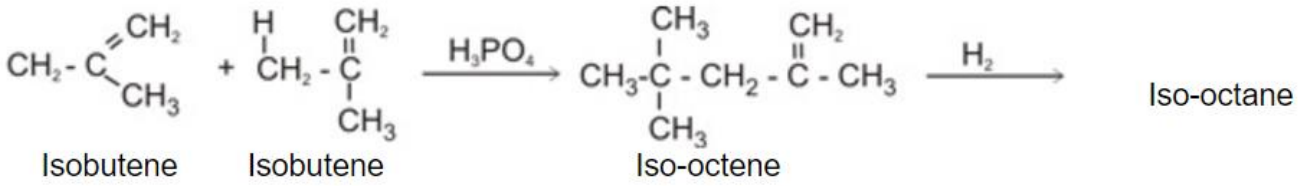
Example, hydrogen is used for the manufacture of ammonia; ethane for ethanol; benzene and propene for the manufacture of drugs, detergents, plastics, synthetic rubber, fertilizers, weed killers and synthetic fibre.

Reformation of Petrol

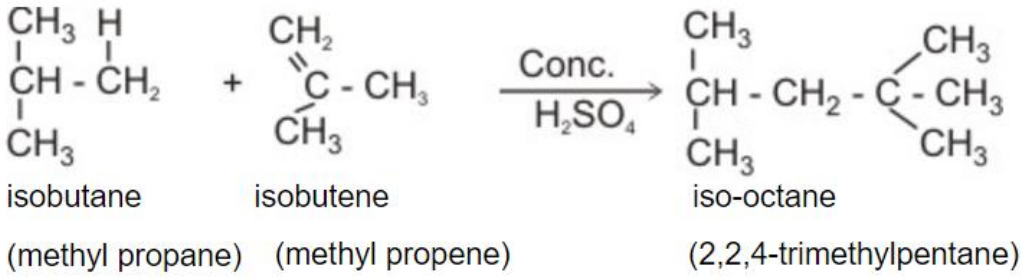
Hydrocarbons in the petrol range, as well as special fuels (e.g. high-octane aviation spirit), can be reformed from gaseous hydrocarbons. The processes which can be used in carrying this out are:

(1) polymerization (2) alkylation (3) isomerization and (4) hydrogenation. These are summarized below:

1. **Polymerization and hydrogenation**



2. **alkylation**



3. **isomerization**



Quality of Petrol and Meaning of Octane Number

The quality of petrol is a measure of how it burns in an internal combustion engine. A poor-quality petrol burns rapidly and unevenly and generates an explosion which disturbs the up and down movement of the piston, resulting in a strange sound normally referred to as 'Knocking' - this petrol contains large proportion of straight chain alkanes.

A good quality petrol on the other hand burns smoothly and evenly - this contains a large proportion of branched alkanes.

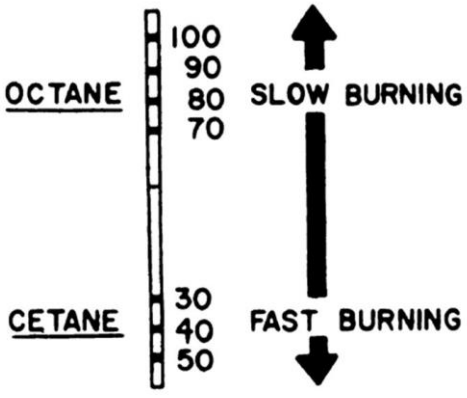
Note: A given petrol consists of both branched and straight chain alkanes.

The **octane number** of a petrol is an arbitrary number which denotes the proportion of the branched-chain to the straight chain alkanes in the petrol. A scale which ranges from zero (for n-hepane - straight chain alkane) to 100 (for 2,2,4 - trimethyl pentane - branched chain alkane) has been devised.

Hence, a petrol with an octane number of 95 consists of 95 parts of the branched alkane (2,2,4 - trimethyl pentane) and just 5 parts of the straight-chain alkane (n-heptane) - this is a very high-quality petrol and is regarded as a 'five star' or 'super' petrol. **The octane rating of the regular petrol is between 85 and 92.**

The main differences between octane and cetane numbers are:

- Octane numbers apply to gasoline and measure the fuel's ability to resist knocking during combustion.
- Cetane numbers apply to diesel fuels and indicate how easily the fuel ignites under compression.
- Octane relates to combustion stability in gasoline engines, while cetane relates to ignition quality in diesel engines.



Octane and Cetane number of fuel

Fuel	Research Octane Number (ROM)	Motor Octane Number (MON)	Cetane number	Boiling point (ºC)
Gasoline	92-98	80-90	0-5	37-205
Diesel	-25	-	45-55	140-360
Methanol	106	92	5	65
Ethanol	107	89	5	79
Propane (LPG)	112	97	-2	-42.15
Methane (CNG)	120	120	0	-161.6

Uses of Petrol Additives

High octane rated petrol is very expensive and might not be readily affordable, especially in the third world countries. Certain chemicals have been developed to be used as additives. These chemicals function to increase the octane rating of low grade petrol by slowing the combustion rate of the petrol - they are therefore called anti-knock.

Examples of petrol additives are tetraethyl lead(IV) (TEL), $Pb(C(C_2H_5)_4$; methanol and ethanol.

Note: Tetraethyl lead(IV) is no longer in use, especially in the developed countries due to its environmental pollution effect - it introduces lead into the environment, thereby causing severe environmental and health hazards.

Which of the following is NOT an example of potential energy?

(1) Chemical energy

(2) Sound energy

(3) Nuclear energy

(4) Gravitational energy

ANSWER: B

Sound energy is NOT an example of potential energy.

Explanation:

- **Chemical Energy**: Stored in the bonds of chemical compounds and released during chemical reactions. It's a form of potential energy.
- **Sound Energy**: Energy carried by sound waves. It's a form of kinetic energy, as it involves the movement of particles.
- **Nuclear Energy**: Stored in the nucleus of atoms and released during nuclear reactions. It's a form of potential energy.
- **Gravitational Energy**: Energy an object possesses because of its position in a gravitational field. It's a form of potential energy.

So, the correct answer is:

(2) Sound energy

Here is a table that summarizes examples of different types of energy, including potential energy, kinetic energy, and other forms:

Type of Energy	Examples	Description
Potential Energy		
Chemical Energy	Batteries, Food, Fossil Fuels	Stored in the bonds of chemical compounds and released during chemical reactions.
Gravitational Energy	A book on a shelf, Water in a dam	Energy an object possesses due to its position in a gravitational field.
Elastic Energy	Stretched rubber band, Compressed spring	Stored in objects that are stretched or compressed.
Nuclear Energy	Atomic nuclei, Nuclear reactors	Stored in the nucleus of atoms and released during nuclear reactions.
Kinetic Energy		
Motion Energy	Moving car, Flying airplane	Energy possessed by an object due to its motion.
Thermal Energy	Boiling water, Sunlight	Energy related to the temperature of an object, due to the movement of its particles.
Sound Energy	Music, Speaking	Energy carried by sound waves, produced by vibrating objects.
Electrical Energy	Flowing electrons, Lightning	Energy carried by moving electrons through a conductor.
Other Forms of Energy		
Radiant Energy	Light, X-rays	Energy carried by electromagnetic waves.
Mechanical Energy	Wind turbine, Clockwork toys	Sum of potential and kinetic energy in an object used to do work.

Arrange the following in decreasing order of their present contribution to total electricity production in India:

(A) Thermal

(B) Hydro

(C) Solar

(D) Nuclear

Choose the correct answer from the options given below:

(1) (A), (B), (C), (D)

(2) (B), (A), (C), (D)

(3) (A), (C), (B), (D)

(4) (C), (A), (D), (B)

ANSWER: C

1.Total Installed Capacity (As on 30.11.2024)- Source : Central Electricity Authority (CEA)

➢ **Installed Generation Capacity (Sectorwise) as on 30.11.2024 :**

Sector	Installed Capacity (MW)	% Share in Total
Central Sector	1,04,453	22.9%
State	1,08,330	23.7%
Private Sector	2,43,974	53.4%
TOTAL	4,56,757	



➢ **Installed Generation Capacity (Fuelwise) as on 30.11.2024 :**

	Category		Installed Generation Capacity (MW)	% Share in Total
Fossil Fuel	Coal		2,11,030	46.2%
	Lignite		6,620	1.4%
	Gas		24,818	5.4%
	Diesel		589	0.1%
	Total Fossil Fuel :		2,43,057	53.2%
Non-Fossil Fuel	RES (Incl. Hydro)		2,05,521	45.0%
	Hydro	46,968		10.3%
	Wind, Solar & Other RE	1,58,552		34.7%
	Wind	*47,959*		*10.5%*
	Solar	*94,168*		*20.6%*
	BM Power/Cogen.	*10,728*		*2.3%*
	Waste to Energy	*613*		*0.1%*
	Small Hydro Power	*5,084*		*1.1%*
	Nuclear		8,180	1.8%
	Total Non-Fossil Fuel :		2,13,701	46.8%
	Total Installed Capacity (Fossil Fuel & Non-Fossil Fuel)		4,56,757	100%

Telegram Channel : Chevish JRF Foundation Whatsapp Number for Inquiry: 8200178822

ALL INDIA INSTALLED CAPACITY (IN MW) OF POWER STATIONS

(As on 30.11.2024)

(UTILITIES)

Region	Ownership/ Sector	Mode wise breakup									Grand Total
		Thermal					Nuclear	Renewable			
		Coal	Lignite	Gas	Diesel	Total		Hydro	RES*(MNRE)	Total	
Northern Region	State	19525.00	250.00	2878.90	0.00	22653.90	0.00	6008.25	815.20	6823.45	29477.35
	Private	22084.33	1080.00	772.00	0.00	23936.33	0.00	3241.00	41873.88	45114.88	69051.21
	Central	15048.62	250.00	2344.06	0.00	17642.68	1620.00	11580.51	379.00	11959.51	31222.19
	Sub Total	**56657.95**	**1580.00**	**5994.96**	**0.00**	**64232.91**	**1620.00**	**20829.76**	**43068.08**	**63897.84**	**129750.75**
Western Region	State	21290.00	900.00	2849.82	0.00	25039.82	0.00	5446.50	604.23	6050.73	31090.55
	Private	31762.17	500.00	4676.00	0.00	36938.17	0.00	481.00	53929.20	54410.20	91348.37
	Central	21610.47	0.00	3280.67	0.00	24891.14	3240.00	1635.00	666.30	2301.30	30432.44
	Sub Total	**74662.64**	**1400.00**	**10806.49**	**0.00**	**86869.13**	**3240.00**	**7562.50**	**55199.73**	**62762.23**	**152871.36**
Southern Region	State	22192.50	0.00	791.98	159.96	23144.44	0.00	11867.48	637.08	12504.56	35649.00
	Private	13572.50	250.00	5120.24	273.70	19216.45	0.00	0.00	56157.34	56157.34	75373.79
	Central	13827.04	3390.00	359.58	0.00	17576.62	3320.00	0.00	541.90	541.90	21438.52
	Sub Total	**49592.04**	**3640.00**	**6271.80**	**433.66**	**59937.50**	**3320.00**	**11867.48**	**57336.32**	**69203.80**	**132461.30**
Eastern Region	State	6970.00	0.00	80.00	0.00	7050.00	0.00	3550.22	278.11	3828.33	10878.33
	Private	5613.00	0.00	0.00	0.00	5613.00	0.00	209.00	2004.85	2213.85	7826.85
	Central	16291.86	0.00	0.00	0.00	16291.86	0.00	1005.20	10.00	1015.20	17307.06
	Sub Total	**28874.86**	**0.00**	**80.00**	**0.00**	**28954.86**	**0.00**	**4764.42**	**2292.96**	**7057.38**	**36012.24**
North Eastern Region	State	0.00	0.00	411.36	36.00	447.36	0.00	422.00	276.25	698.25	1145.61
	Private	0.00	0.00	0.00	0.00	0.00	0.00	0.00	308.94	308.94	308.94
	Central	1242.02	0.00	1253.60	0.00	2495.62	0.00	1522.01	30.00	1552.01	4047.63
	Sub Total	**1242.02**	**0.00**	**1664.96**	**36.00**	**2942.98**	**0.00**	**1944.01**	**615.19**	**2559.20**	**5502.18**
Islands	State	0.00	0.00	0.00	84.35	84.35	0.00	0.00	5.25	5.25	89.60
	Private	0.00	0.00	0.00	35.19	35.19	0.00	0.00	29.78	29.78	64.97
	Central	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.10	5.10	5.10
	Sub Total	**0.00**	**0.00**	**0.00**	**119.54**	**119.54**	**0.00**	**0.00**	**40.13**	**40.13**	**159.67**
ALL INDIA	State	69977.50	1150.00	7012.06	280.31	78419.87	0.00	27294.45	2616.12	29910.57	108330.44
	Private	73032.00	1830.00	10568.24	308.89	85739.14	0.00	3931.00	154303.99	158234.99	243974.13
	Central	68020.00	3640.00	7237.91	0.00	78897.91	8180.00	15742.72	1632.30	17375.02	104452.93
	Total	**211029.50**	**6620.00**	**24818.21**	**589.20**	**243056.91**	**8180.00**	**46968.17**	**158552.41**	**205520.58**	**456757.49**

Figures at decimal may not tally due to rounding off

Abbreviation:- SHP=Small Hydro Project (≤ 25 MW), BP=Biomass Power, U&I=Urban & Industrial Waste Power, RES=Renewable Energy Sources

Note : - 1. RES include SHP, BP, U&I, Solar and Wind Energy. Installed capacity in respect of RES (MNRE) as on 30.11.2024 (As per latest information available with MNRE)

***Break up of RES all India as on 30.11.2024 is given below (in MW) :**

Small Hydro Power	Wind Power	Bio-Power		Solar Power[$]	Total Capacity
		BM Power/Cogen.	Waste to Energy[#]		
5084.25	47959.13	10728.21	612.99	94167.83	158552.41

[#] Includes Waste to Energy and Waste to Energy (Off-grid)

[$] Includes Ground Mounted Solar, Rooftop Solar, Hybrid Solar Comp. and Off-grid Solar/ KUSUM

Ethanol blended in proportion of 10% with gasoline is called :

(1) Biodiesel

(2) Synthetic Natural Gas

(3) Hydrogen fuel

(4) Gasohol

ANSWER: D

Ethanol blended in proportion of 10% with gasoline is called **Gasohol**.

Gasohol is a blend of gasoline and ethanol, where ethanol can be used in varying proportions. The common blend of 10% ethanol with 90% gasoline is specifically known as E10.

Here is a table summarizing different fuel names and their corresponding blends with biofuels:

Fuel Name	Blend Composition	Description
Gasohol (E10)	10% Ethanol, 90% Gasoline	Common blend of ethanol and gasoline, used to reduce emissions and improve octane rating.
E85	85% Ethanol, 15% Gasoline	Higher ethanol blend, used in flexible-fuel vehicles (FFVs) designed to run on higher ethanol content.
Biodiesel (B20)	20% Biodiesel, 80% Diesel	Blend of biodiesel and conventional diesel fuel, used to reduce emissions and improve lubricity.
B100 (Pure Biodiesel)	100% Biodiesel	Pure biodiesel, made from vegetable oils or animal fats, used as a renewable alternative to diesel fuel.

JUNE 2024 Cancelled

Telegram Channel : Chevish JRF Foundation Whatsapp Number for Inquiry: 8200178822

The peak of coal bed methane generation during catagenesis occur at a temperature of:

(1) 20° C

(2) 50° C

(3)150° C

(4) 200° C

ANSWER: C

What is Coal Bed Methane (CBM)?

Coal bed methane (CBM) is a form of natural gas extracted from coal beds. It is predominantly composed of methane (CH_4) and is found in coal seams, where it is adsorbed onto the surface of coal particles.

What is Catagenesis?

Catagenesis is the middle phase of the transformation of organic matter into hydrocarbons. This process occurs at deeper burial depths and higher temperatures compared to diagenesis, the initial transformation stage. During catagenesis, the thermal breakdown of organic material (kerogen) in sedimentary rocks results in the generation of hydrocarbons, including methane.

Temperature and Methane Generation

The peak of coal bed methane generation occurs at a specific temperature range during catagenesis. Let's break this down:

1. **Temperature Range**:
    - Methane generation from coal is highly dependent on temperature.
    - During catagenesis, temperatures typically range from 60°C to 200°C.
    - The peak generation of methane, however, is observed at around 150°C.

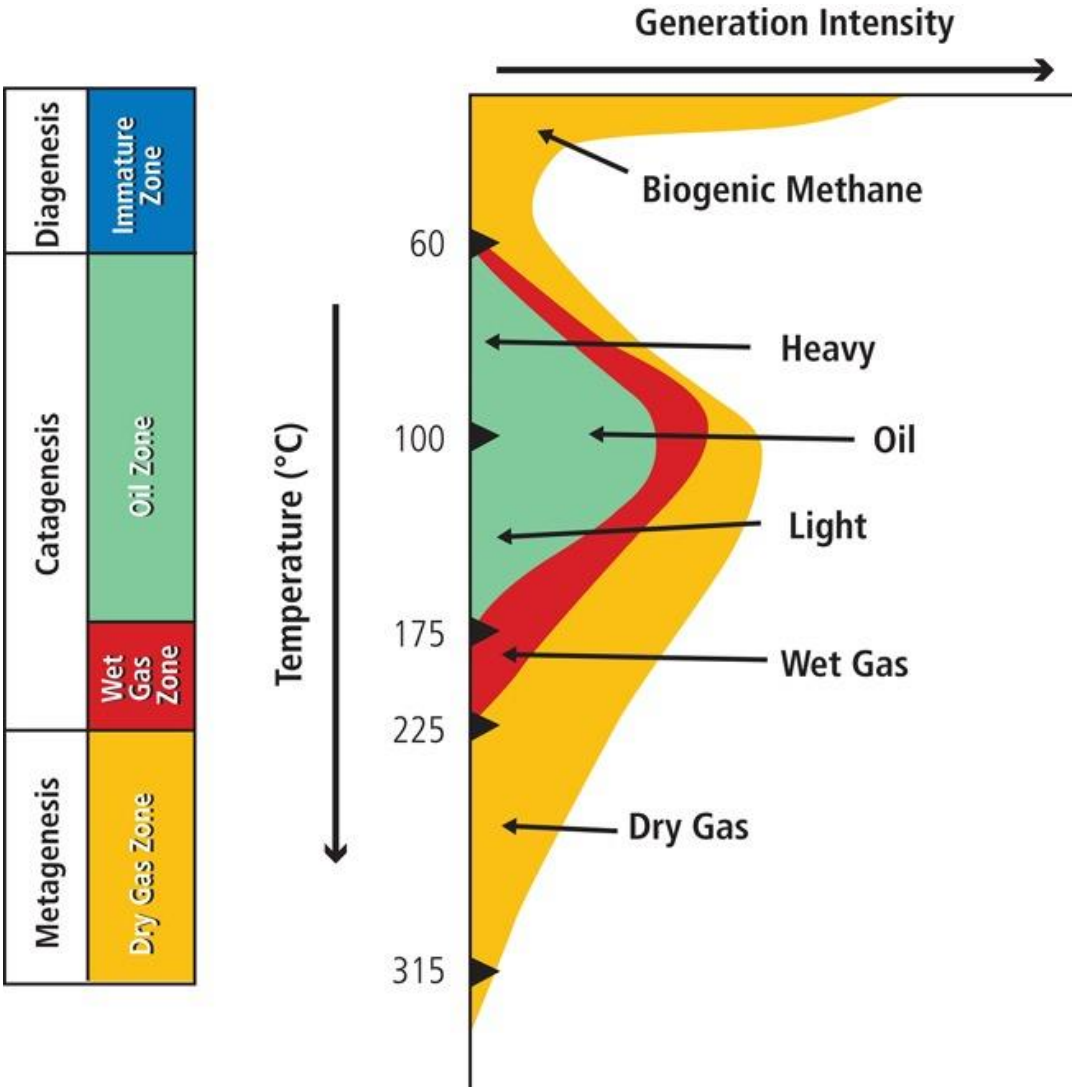
2. **Thermal Maturation**:
    - As temperature increases, organic matter in coal undergoes thermal maturation.
    - During this process, complex organic molecules are broken down into simpler hydrocarbons.
    - At around 150°C, the breakdown of these organic molecules reaches an optimal point, resulting in the maximum release of methane.
3. **Chemical Reactions**:
    - The thermal breakdown (thermal cracking) of kerogen and other organic matter in coal releases methane.
    - This process involves breaking the carbon-hydrogen bonds in organic molecules, releasing methane gas.

Importance of Methane Generation Temperature

Understanding the temperature at which methane generation peaks is crucial for several reasons:

1. **Exploration and Extraction**:
    - Identifying regions where coal seams have been exposed to the optimal temperature range (around 150°C) helps in locating potential CBM resources.
    - This information aids in efficient extraction and management of CBM.
2. **Safety and Environmental Impact**:
    - CBM extraction needs to be managed carefully to prevent uncontrolled release of methane, which is a potent greenhouse gas.
    - Understanding the temperature-related generation of methane can help in implementing safety measures and minimizing environmental impact.
3. **Energy Production**:
    - CBM is a valuable source of clean energy.

- Knowing the temperature conditions for peak methane generation can optimize extraction processes, ensuring a steady supply of natural gas.



Which of the following coalfields contain tertiary coals?

A. Raniganj coalfields, West Bengal

B. Neyveli coalfields, Tamil Nadu

C. Talchir coalfields, Odisha

D. Makum coalfields, Assam

Choose the correct answer from the options given below:

(1) A, B, C only

(2) B, C, D only

(3) A and C only

(4) B and D only

ANSWER: D

Tertiary Coals

Tertiary coals are younger than Gondwana coals, having formed during the Tertiary period (65 to 2.6 million years ago). These coals are typically found in regions that have undergone significant geological changes since the end of the Mesozoic era. Tertiary coals often include lignite (brown coal), which is softer and has a lower carbon content compared to bituminous coal found in older geological formations.

Key Coalfields with Tertiary Coals

1. Neyveli Coalfields, Tamil Nadu:

- **Location**: Neyveli, in the Cuddalore district of Tamil Nadu.
- **Type of Coal**: Lignite (brown coal), which is a type of Tertiary coal.

- **Features**: These coalfields are renowned for their significant lignite deposits. Lignite is characterized by its high moisture content and is used primarily for electricity generation in thermal power plants.
- **Importance**: Neyveli coalfields are crucial for the energy sector in Tamil Nadu and India, supplying lignite for various industrial purposes, including power generation and chemical manufacturing.

2. Makum Coalfields, Assam:

- **Location**: Near the town of Margherita in the Tinsukia district of Assam.
- **Type of Coal**: Tertiary coals, including high-grade bituminous coal and semi-anthracite.
- **Features**: The coal found in Makum coalfields is known for its high carbon content and calorific value, making it suitable for industrial applications such as steel production and energy generation.
- **Importance**: These coalfields are a major source of high-quality coal in the northeastern region of India and play a significant role in the regional economy.

Other Coalfields Mentioned

1. Raniganj Coalfields, West Bengal:

- **Location**: Asansol-Durgapur region in West Bengal.
- **Type of Coal**: Gondwana coals, primarily bituminous coal.
- **Features**: Raniganj coalfields are one of the oldest and largest coal mining areas in India, known for high-quality bituminous coal suitable for both coking and non-coking purposes.

2. Talchir Coalfields, Odisha:

- **Location**: Talcher in the Angul district of Odisha.
- **Type of Coal**: Gondwana coals, including bituminous and sub-bituminous coal.
- **Features**: These coalfields are significant for their extensive reserves of bituminous coal, used for thermal power generation and industrial applications.

Distribution of Coal in India

- **Gondwana coal fields [250 million years old]**
- **Tertiary coal fields [15 – 60 million years old]**

Gondwana Coal

- Gondwana coal makes up to **98 per cent of the total reserves and 99 per cent of the production of coal in India.** Satpuras, denudation [weathering + erosion] has exposed coal bearing Gondwana strata.
- The carbon content in Gondwana coal **[250 million years old]** is less compared to the Carboniferous coal **[350 million years old][Almost Absent in India]** because of its much younger age.
- Gondwana coal forms India's metallurgical grade as well as superior quality coal.
- The **Damuda series (i.e. Lower Gondwana)** possesses the best worked coalfields accounting for 80 per cent of the total coal production in India. 80 out of 113 Indian coalfields are located in the rock systems of the **Damuda series [**lower Gondwana Age**]**.
- Coking as well as non-coking and bituminous as well as sub-bituminous coal are obtained from Gondwana coal fields.
- **Anthracite** is generally not found in the Gondwana coal fields.
- The volatile compounds and ash (usually 13 – 30 per cent) and doesn't allow Carbon percentage to rise above **55 to 60 per cent.** [It requires few million years more if the quality has to get better. Remember Gondwana coal is 100 million years younger than Carboniferous coal].
- Gondwana coal is free from moisture, but it contains **Sulphur** and **Phosphorous.**
- These basins occur in the valleys of certain rivers viz., the Damodar (Jharkhand-West Bengal); the Mahanadi (Chhattisgarh-Odisha); the Son (Madhya Pradesh Jharkhand); the Godavari and the Wardha (Maharashtra-Andhra Pradesh); the Indravati, the Narmada, the Koel, the Panch, the Kanhan and many more.

Distribution of Gondwana Coal in India

- First coal mine was opened in 1774 at **Raniganj in West Bengal.**
- Coal industry was nationalized in 1973-74. [The present government made some serious changes during the last year [2015] by allowing private sector to play a bigger role in coal production].
- India is now the **third largest coal producer** in the world after **China and the USA**.
- Coal industry provides employment to nearly seven lakh persons.
- Gondwana Coalfields == exclusively found in the Peninsular plateau of India.

Tertiary Coal

- Tertiary coal 15 to 60 million years old. Carbon content is very low.
- Mainly confined to the extra-Peninsula [Jammu and Kashmir, Himachal Pradesh, Assam, Arunachal Pradesh etc.]
- Coal generally has **low carbon** and **high percentage of moisture** and **Sulphur**.[It takes few hundred million years for the carbon content to improve].
- Important areas of Tertiary coal include parts of Assam, Meghalaya, Arunachal Pradesh, Nagaland, Himalayan foothills of Darjeeling in West Bengal, Jammu and Kashmir, Uttar Pradesh, Rajasthan, Kerala,
- Tamil Nadu and the union territory of Pondicherry also bear tertiary coal reserves [exceptions].

Tertiary Coalfields in Assam

- Makum, Nazira, Mikir Hills, Dilli-Jeypore and Lakhuni.
- **Makum coalfield** in Sibsagar district is the most developed field.
- Assam coals contain very **low ash and high coking qualities** but the **sulphur content is high**, as a result of which this coal is not suitable for metallurgical purposes.

- But these coals are best suited for **hydrogenation process** and are used for **making liquid fuels.**

Coal Reserves In India By State

Name of the state	Reserves in billion tonne	% of total reserves
1. JHARKHAND	80.71	26.76
2. ODISHA	75.07	24.89
3. CHATTISHGARH	52.53	17.42
4. WEST BENGAL	31.31	10.38
5. MADHYA PRADESH	25.67	8.51
6. ANDHRA PRADESH	22.48	7.45
7. MAHARASTRA	10.98	3.64
8. OTHERS	2.81	0.95

Match List I with List - II.

List - I Energy resources	List - II Age of associated rock formations
A. Nuclear minerals	I. Recent
B. Coal	II. Paleogene
C. Petroleum	III. Permian
D. Gas hydrates	IV. Precambrian

Choose the correct answer from the options given below :

(1) A-III, B-III, C-IV, D-I

(2) A-IV, B-III, C-II, D-I

(3) A-II, B-I, C-IV, D-III

(4) A-I, B-II, C-III, D-IV

ANSWER: B

A. Nuclear minerals - Precambrian

Nuclear minerals, such as uranium and thorium, are often found in Precambrian rocks. The Precambrian is the earliest part of Earth's history, spanning from the formation of the Earth around 4.6 billion years ago to approximately 541 million years ago. During this time, the Earth's crust was forming and solidifying, creating large deposits of various minerals, including those used for nuclear energy.

B. Coal - Permian

Coal is a sedimentary rock that forms from the accumulation and burial of plant material in ancient swamps and peat bogs. The Permian period, which occurred

about 299 to 252 million years ago, was a time when vast swampy forests existed. These environments provided the perfect conditions for the formation of coal. Over millions of years, the plant material was buried and subjected to heat and pressure, transforming it into coal.

C. Petroleum - Paleogene

Petroleum, also known as crude oil, is formed from the remains of ancient marine organisms such as plankton and algae. The Paleogene period, which spans from about 66 to 23 million years ago, is significant for the formation of many petroleum reserves. During this time, the Earth's climate was warmer, and large amounts of organic material were deposited in marine environments. Over time, this material was buried, and heat and pressure transformed it into petroleum.

D. Gas hydrates - Recent

Gas hydrates are crystalline solids formed from water and gas, typically methane. They are found in marine sediments and permafrost regions. The "Recent" period refers to the current geological epoch, the Holocene, which began approximately 11,700 years ago and continues to the present day. Gas hydrates are most commonly associated with recent geological formations in deep ocean sediments and cold environments where the conditions are right for their formation.

Correct Answer:

So, based on the detailed explanations above, the correct match is:

- **A-IV (Nuclear minerals - Precambrian)**
- **B-III (Coal - Permian)**
- **C-II (Petroleum - Paleogene)**
- **D-I (Gas hydrates - Recent)**

The correct answer is option **(2)**.

Arrange the following coal ranks in increasing order of atomic O/C (oxygen/carbon) ratios:

A Peat

B. Lignite

C. Sub-bituminous

D. Low volatile bituminous

E. Anthracite

Choose the correct answer from the options given below:

(1) A, B, C, D, E

(2) B, C, D, E, A

(3) E, D, C, B, A

(4) A, E, B, C, D

Answer: D

Let's arrange the coal ranks based on their increasing atomic O/C ratios:

1. **Peat**: Peat has a relatively low carbon content and a low O/C ratio.
2. **Lignite**: Lignite has a slightly higher carbon content than peat, but still a relatively low O/C ratio.
3. **Sub-bituminous**: Sub-bituminous coal has a higher carbon content and a higher O/C ratio compared to lignite.
4. **Low volatile bituminous**: Low volatile bituminous coal has an even higher carbon content and a higher O/C ratio.
5. **Anthracite**: Anthracite coal has the highest carbon content and the highest O/C ratio among these ranks.

Therefore, the correct order is **(4) A, E, B, C, D**. Anthracite has the highest atomic O/C ratio, followed by low volatile bituminous, sub-bituminous, lignite, and finally peat

Arrange the following in their ascending order of Lower Heating Values (LHV):

A. Anthracite

D. Crude oil

B. Bituminous

E. Methane

C. Lignite

Choose the correct answer from the options given below:

(1) A, C, B, E, D

(2) B. C, A, D, E

(3) C, B, A, D, E

(4) D, E, C, B, A

ANSWER: C

Let's arrange the fuels in ascending order of their Lower Heating Values (LHV). The LHV represents the heat released when a specified quantity of fuel is combusted, and the latent heat of vaporization of water in the reaction products is **not** recovered. Here's the correct order:

1. **Lignite (C)**: Lignite has the lowest LHV due to its high moisture content and low heating value.
2. **Bituminous (B)**: Bituminous coal has a higher LHV than lignite.
3. **Crude oil (D)**: Crude oil has a higher LHV than bituminous coal.
4. **Methane (E)**: Methane (natural gas) has a higher LHV than crude oil.
5. **Anthracite (A)**: Anthracite coal has the highest LHV among the listed fuels.

Therefore, the correct order is **(3) C, B, A, D, E**.

TABLE A-7E Properties of Some Common Fuels and Hydrocarbons

Fuel (Phase)	Formula	Molar Mass, lbm/lbmol	Density,* lbm/ft³	Enthalpy of Vaporization,† Btu/lbm	Specific Heat,* c_p Btu/lbm·°F	Higher Heating Value,‡ Btu/lbm	Lower Heating Value,‡ Btu/lbm
Carbon (*s*)	C	12.011	125	—	0.169	14,100	14,100
Hydrogen (*g*)	H_2	2.016	—	—	3.44	60,970	51,600
Carbon monoxide (*g*)	CO	28.013	—	—	0.251	4,340	4,340
Methane (*g*)	CH_4	16.043	—	219	0.525	23,880	21,520
Methanol (ℓ)	CH_4O	32.042	49.3	502	0.604	9,740	8,570
Acetylene (*g*)	C_2H_2	26.038	—	—	0.404	21,490	20,760
Ethane (*g*)	C_2H_6	30.070	—	74	0.418	22,320	20,430
Ethanol (ℓ)	C_2H_6O	46.069	49.3	395	0.583	12,760	11,530
Propane (ℓ)	C_3H_8	44.097	31.2	144	0.662	21,640	19,930
Butane (ℓ)	C_4H_{10}	58.123	36.1	156	0.578	21,130	19,510
1-Pentene (ℓ)	C_5H_{10}	70.134	40.0	156	0.525	20,540	19,190
Isopentane (ℓ)	C_5H_{12}	72.150	39.1	—	0.554	20,890	19,310
Benzene (ℓ)	C_6H_6	78.114	54.7	186	0.411	17,970	17,240
Hexene (ℓ)	C_6H_{12}	84.161	42.0	169	0.439	20,430	19,090
Hexane (ℓ)	C_6H_{14}	86.177	41.2	157	0.542	20,770	19,240
Toluene (ℓ)	C_7H_8	92.141	54.1	177	0.408	18,230	17,420
Heptane (ℓ)	C_7H_{16}	100.204	42.7	157	0.535	20,680	19,180
Octane (ℓ)	C_8H_{18}	114.231	43.9	156	0.533	20,590	19,100
Decane (ℓ)	$C_{10}H_{22}$	142.285	45.6	155	0.528	20,490	19,020
Gasoline (ℓ)	$C_nH_{1.87n}$	100–110	45–49	151	0.57	20,300	18,900
Light diesel (ℓ)	$C_nH_{1.8n}$	170	49–52	116	0.53	19,800	18,600
Heavy diesel (ℓ)	$C_nH_{1.7n}$	200	51–55	99	0.45	19,600	18,400
Natural gas (*g*)	$C_nH_{3.8n}N_{0.1n}$	18	—	—	0.48	21,500	19,200

知乎 @Simultaneous

Telegram Channel : Chevish JRF Foundation Whatsapp Number for Inquiry: 8200178822

Match List - 1 with List - II.

List-1 Coal type	List - II Deposit
A. Coal	1. Clathrates
B. Petroleum	II. Seam
C. Nuclear Minerals.	III. Trap
D. Gas hydrates	IV. Placers

Choose the correct answer from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-1, B-II, C-III, D-IV

(3) A-II, B- III, C-IV, D-I

(4) A-III, B-II, C-IV, D-I

ANSWER: C

Let's match the coal types with their corresponding deposits:

A. Coal

- **II. Seam**: Coal is typically found in seams or layers within sedimentary rocks.

B. Petroleum

- **III. Trap**: Petroleum accumulates in traps formed by geological structures that prevent its migration, such as anticlines, faults, and salt domes.

C. Nuclear Minerals

- **IV. Placers**: Certain nuclear minerals, such as uranium and thorium, can be found in placer deposits, which are formed by the concentration of heavy minerals by water.

D. Gas hydrates

- **I. Clathrates**: Gas hydrates, also known as clathrates, are crystalline water-based solids in which gas molecules are trapped within a lattice of water ice.

Summary of Matching:

Coal Type	Deposit
A. Coal	**II. Seam**
B. Petroleum	**III. Trap**
C. Nuclear Minerals	**IV. Placers**
D. Gas hydrates	**I. Clathrates**

Correct Answer:

3. A-II, B-III, C-IV, D-I

Arrange the following stages of formation of petroleum deposits from initial to last:

A. Migration

B. Seepage

C. Generation

D. Accumulation

Choose the correct answer from the options given below:

(1) A, B, C, D

(2) D, C, A, B

(3) C, A, D, B

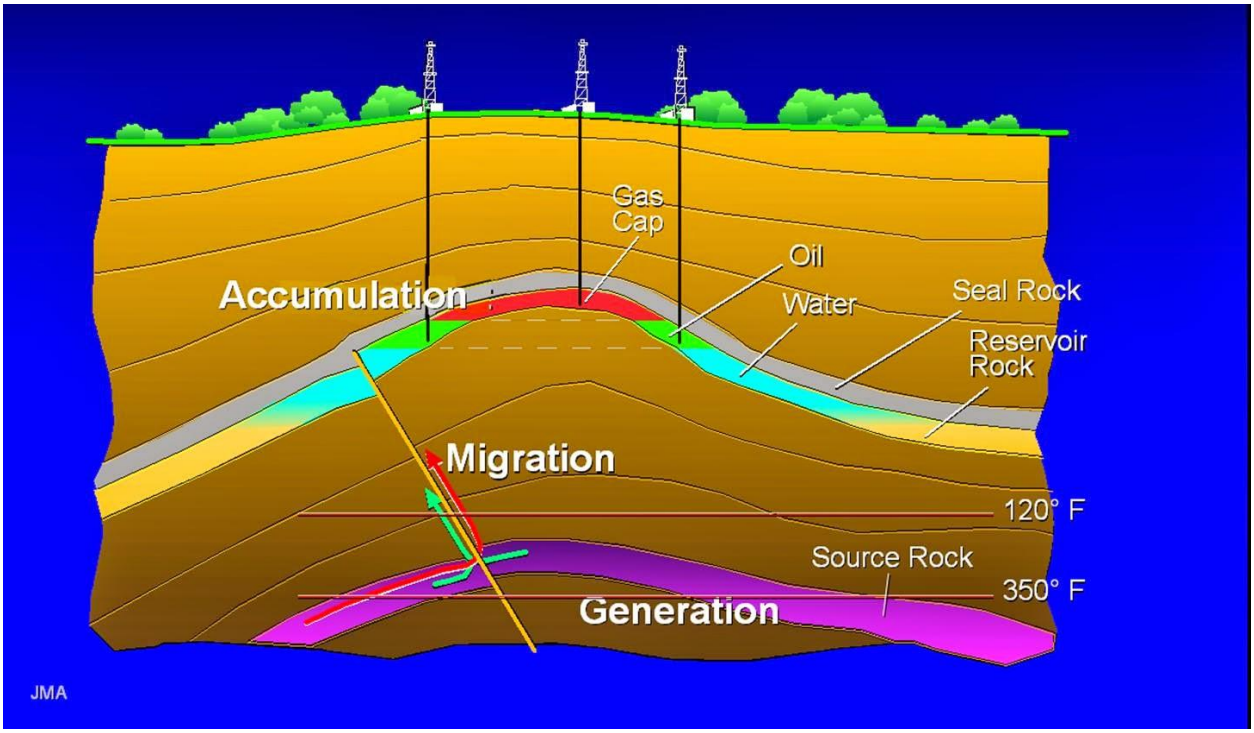
(4) C, A, B, D

Answer: C

The stages of petroleum deposit formation occur over millions of years. Let's arrange them in order:

1. **Generation (C)** : Oil generation begins with productive source beds rich in organic carbon content. These beds undergo three stages of transformation: diagenesis, metagenesis, and catagenesis
2. **Migration (A)**: After oil forms, it migrates from the source rock to a porous sedimentary reservoir rock.
3. **Accumulation (D)**: The migrating hydrocarbons accumulate in traps within the reservoir rock.
4. **Seepage (B)**: Seepage refers to the release of oil or gas to the surface, but it's not a distinct stage in the formation process.

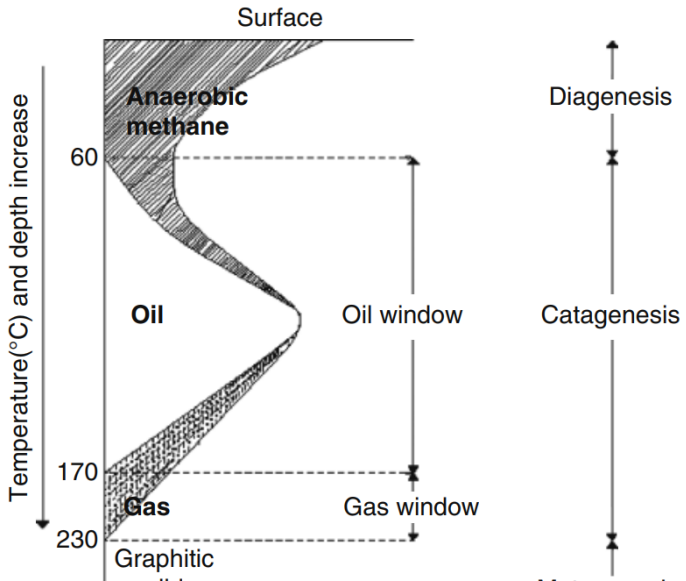
Therefore, the correct order is **(3) C, A, D, B**.



FORMATION OF FOSSIL FUELS

Diagenesis: from organic matter to kerogen

Catagenesis: from kerogen to fossil fuels



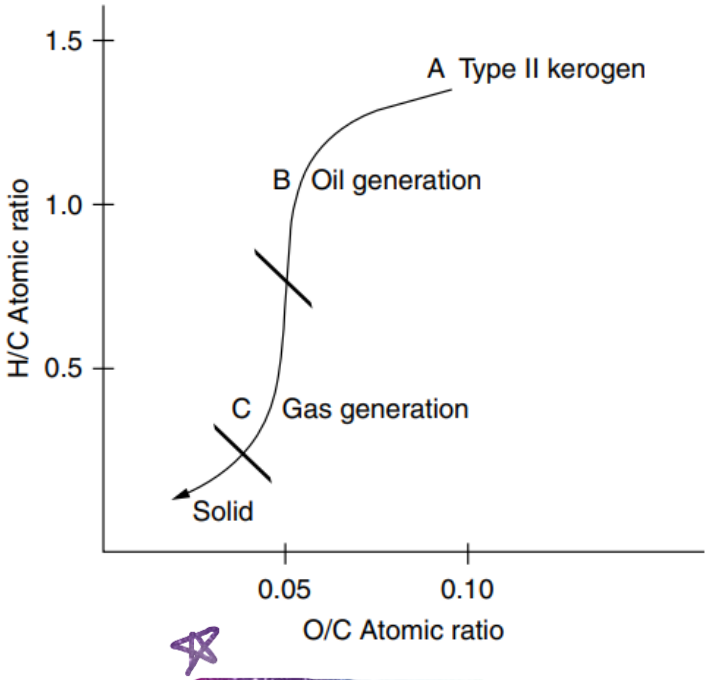


Figure 8.7 The van Krevelen diagram allows tracking of compositional changes in the catagensis of kerogen, as shown here for Type II kerogen. This type of diagram complements Figure 8.6.

Telegram Channel : Chevish JRF Foundation Whatsapp Number for Inquiry: 8200178822